TOPIC 3 LAW OF LARGE NUMBERS (LLN)

Name: ____________________ Student No.: ____________ Course/Section: ____________ Date: __________

LEARNING OBJECTIVES

After completing this topic, students should be able to:

1. Define the Law of Large Numbers (LLN).
2. Explain how the sample mean gets closer to the population mean as sample size increases.
3. Identify the effect of different sample sizes on the accuracy of the sample mean.
4. Solve simple problems that illustrate the LLN.

As sample size increases, the sample mean gets closer to the population mean μ.

A. CONCEPT CHECK

Answer the following questions.

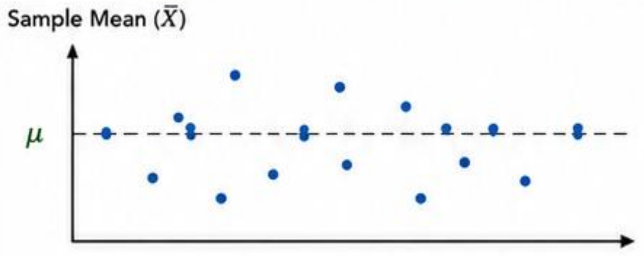
1. What does the Law of Large Numbers (LLN) state? ________________________________
2. As the sample size increases, the sample mean gets closer to which value? ____________________
3. The symbol for the population mean is __________.
4. The symbol for the sample mean based on n observations is __________.
5. Fill in the blanks to complete the statement of LLN.

As the sample size increases ($n \to \infty$), the sample mean $\overline{X}_n$ tends to the population mean μ.

B. SEE THE IDEA

Look at the two situations below. Each dot represents the sample mean from one random sample taken from the same population with true mean μ (shown by the dashed line).

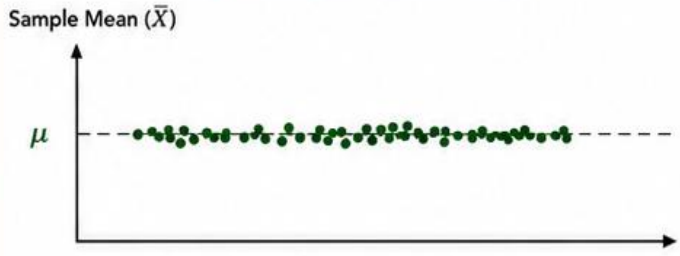
Small Sample Size (n = 5)



Each dot is the mean of 5 observations.

- Sample means are more scattered.
- Averages may be less accurate.

Large Sample Size (n = 100)



Each dot is the mean of 100 observations.

- Sample means are very close to μ.
- Averages are more accurate.

1. Based on the graphs, which sample size gives more scattered sample means? Why?

____

____

2. Which sample size gives sample means that are closer to μ? Why?

____

____

3. What do the graphs suggest about the sample mean as the sample size increases?

____

____

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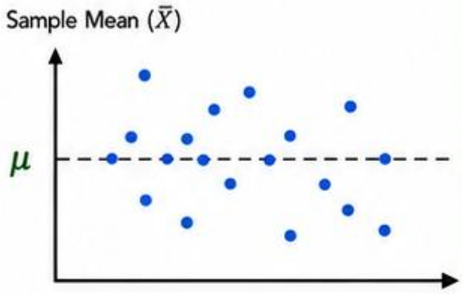
C. OBSERVE THE PATTERN

Observe the sampling distributions below. Each dot represents the sample mean from one random sample taken from the same population with true mean μ. (shown by the dashed line).

Small Sample Size (n = 5)

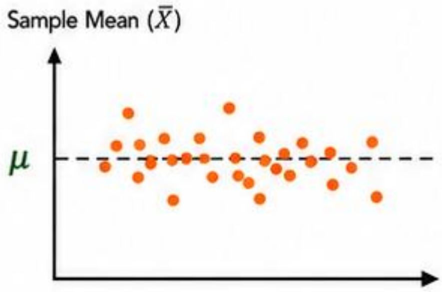


- Sample means are **widely spread out.**
- Less consistent estimates of μ.

Interpretation:

1. Variability (spread of sample means): ________
   (High / Moderate / Low)
2. Accuracy (closeness to μ): ____________
   (High / Moderate / Low)

Medium Sample Size (n = 30)

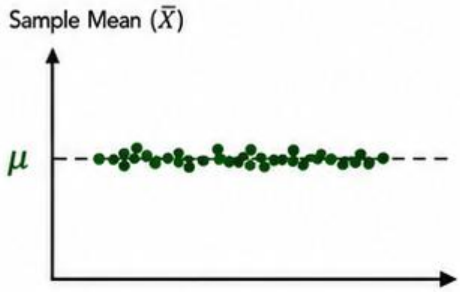


- Sample means are moderately spread out.
- More consistent estimates of μ.

Interpretation:

1. Variability (spread of sample means): ________
   (High / Moderate / Low)
2. Accuracy (closeness to μ): ____________
   (High / Moderate / Low)

Large Sample Size (n = 100)



- Sample means are **tightly clustered.**
- Very consistent estimates of μ.

Interpretation:

1. Variability (spread of sample means): ________
   (High / Moderate / Low)
2. Accuracy (closeness to μ): ____________
   (High / Moderate / Low)

Compare the three situations above.

Sample Size (n)	Spread of Sample Means (Variability)	Closeness to μ (Accuracy)	Overall Description
5	High variability (sample means are widely spread out.)		
30		Moderate accuracy (sample means are moderately close to μ.)	
100			Sample means are highly stable and very close to μ. (very consistent estimates.)

1. As the sample size increases from 5 to 30 to 100, what happens to the variability (spread) of the sample means?

____

____

2. As the sample size increases, what happens to the closeness of the sample means to the population mean μ?

____

____

3. What overall conclusion about the sample mean can be drawn from the patterns above?

____

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Fill in the blanks with the correct words from the box.

population mean (μ)	closer	larger	more accurate	stable	Law of Large Numbers

1. The ____________________ states that as the sample size increases, the sample mean becomes closer to the population mean.
2. When the sample size is small, the sample mean is ____________________ and less accurate.
3. When the sample size is large, the sample mean is ____________________ and more accurate.
4. The value the sample mean approaches is the ____________________.
5. Larger samples provide estimates that are more ____________________ and reliable.

The true average test score of all students in a university is $\mu = 75$.
Three different random samples are taken from the same population.
Compute the sample mean for each sample and the difference from μ.

Sample Size (n)	Sample Scores	Sample Mean ($\overline{X}$)	Difference $\lvert\overline{X} - \mu\rvert$
5	68, 74, 80, 72, 78		
30	73, 76, 78, 74, 77, 72, 75, 74, 76, 77, 73, 75, 74, 76, 75, 77, 72, 74, 75, 76, 73, 74, 75, 76, 73, 74, 75, 77, 76, 74		
100	(100 scores collected)		

1. What do you notice about the sample mean and its difference from μ as the sample size increases?

____

____

2. Based on this example, what can you conclude about the Law of Large Numbers?

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